

DAY 16 & 17: HYPOTHESIS TESTING & P-VALUE

Data Analysis Mathematics Series — Deciding "By Chance" vs "Real Impact"

SECTION 1: CORE THEORY / MAIN CONCEPT

1. Hypothesis Testing Formulation

Null Hypothesis (H_0): The status quo framework. It states that there is no statistical significance, no true difference, or no effect. Any observed change is due to random variance.

Null Hypothesis (H_0): Yeh purani soch ya default position hoti hai jo kehti hai ki data me koi asli badlav nahi aaya hai, sab pehle jaisa normal chal raha hai. Kisi bhi variation ko ye kismat (random chance) manti hai. Isme hamesha Equal (=) sign aata hai.

Alternative Hypothesis (H_1 or H_a): The directional or non-directional claim being investigated. It asserts a true, systematic effect or operational shift exists in the population.

Alternative Hypothesis (H_1): Yeh naya claim ya daawa hota hai jo kehta hai ki data me sach me koi bada badlav ya effect aaya hai. Isme hamesha Not Equal ($\neq$), Less than ($<$), ya Greater than ($>$) ka sign hota hai.

2. The p-Value Mechanics ("By Chance" vs "Real")

p-Value (Probability Value): The statistical probability of obtaining a sample result at least as extreme as the observed metrics, assuming the Null Hypothesis is true.

- **High p-value:** Result is highly probable under H_0 (Likely "By Chance").
- **Low p-value:** Result is highly improbable under H_0 (Likely "Real Operational Shift").

p-Value: Yeh probability batati hai ki agar Null Hypothesis sach hoti, to kismat ke bharose aisa extreme result aane ka kitna chance tha.

- **High p-value:** Yani kismat se aisa data milna aam baat hai ("By Chance" hua hai).
- **Low p-value:** Yani kismat se aisa data aana lagbhag impossible hai ("Real Impact" hua hai).

SECTION 2: SIGNIFICANCE LEVELS & DECISION BOUNDARIES

⚠ SCENARIO A: p-Value > Alpha (α) — Fail to Reject Null

Condition: Calculated p-value exceeds the threshold alpha level (typically 0.05).

Statistical Action: Fail to Reject H_0 . The proof is statistically insufficient to validate the new operational claim.

Condition: Calculated p-value jab threshold alpha level (5%) se bada hota hai.

Action: Fail to Reject H_0 . Hum bolenge ki saboot kamzor hai, badlav real nahi lag raha, yeh naturally bhi ho sakta tha.

✅ SCENARIO B: p-Value $\leq$ Alpha (α) — Reject Null

Condition: Calculated p-value drops below or equals the alpha boundary line (0.05).

Statistical Action: Reject H_0 . The dataset contains overwhelming proof favoring the alternative operational paradigm.

Condition: Calculated p-value jab boundary alpha level (5%) se chota ya uske barabar hota hai.

Action: REJECT NULL HYPOTHESIS (H_0). Hum danka baja kar bolte hain ki data me pakka saboot hai, yeh badlav kismat se nahi balki real impact se hua hai.

SECTION 3: MATHEMATICAL NUMERICAL TESTS (116 VS 120 PROOF)

Core Mathematical Metrics:

$$\text{Z-Score Vector Formula: } Z = (x - \mu) / \sigma$$

Case Study Problem Setup

A jewelry unit maintains a historical baseline population mean of $\mu = 100$ rings/month with a baseline standard deviation of $\sigma = 10$. A new machine is trialed under Alpha $\alpha = 0.05$.

Test 1 Score (x) = 116 rings | Test 2 Score (x) = 120 rings

Gorakhpur jewelry factory ka historical production mean $\mu = 100$ rings hai aur standard deviation $\sigma = 10$ hai. Nayi machine ka test Alpha $\alpha = 0.05$ par do alag target outcomes ke liye prove karen:

Test 1 Score (x) = 116 rings | Test 2 Score (x) = 120 rings

Step-by-Step Computational Solutions

Execution 1 (Production Score = 116):

1. $Z = (116 - 100) / 10 = 16 / 10 = 1.6$
2. Cumulative Area under $Z=1.6$ is 0.9452. Leftover tail probability: **p-value** = $1 - 0.9452 = 0.0548$ (5.48%).
3. **Decision:** Since $0.0548 > 0.05$, we **Fail to Reject H_0** . The 116 outcome could naturally happen by chance.

Calculation 1 (Production Score = 116):

1. $Z\text{-score} = (116 - 100) / 10 = 16 / 10 = 1.6$
2. Z-Table se $Z=1.6$ ka back area 0.9452 hota hai. Top tail probability: **p-value** = $1 - 0.9452 = 0.0548$ (5.48%).
3. **Faisla:** Kyunki $5.48\% > 5\%$ (α), hum ****Null Hypothesis Reject nahi kar sakte****. Yeh kismat se bhi ho sakta tha.

Execution 2 (Production Score = 120):

1. $Z = (120 - 100) / 10 = 20 / 10 = 2.0$
2. Cumulative Area under $Z=2.0$ is 0.9772. Leftover tail probability: **p-value** = $1 - 0.9772 = 0.0228$ (2.28%).
3. **Decision:** Since $0.0228 \leq 0.05$, we **REJECT H_0** . The operational change is statistically real!

Calculation 2 (Production Score = 120):

1. $Z\text{-score} = (120 - 100) / 10 = 20 / 10 = 2.0$
2. Z-Table se $Z=2.0$ ka back area 0.9772 hota hai. Top tail probability: **p-value** = $1 - 0.9772 = 0.0228$ (2.28%).
3. **Faisla:** Kyunki $2.28\% \leq 5\%$ (α), hum ****Null Hypothesis ko REJECT karte hain****. Yeh badlav sach me real machine ka asar hai!